School of Future Tech

Case Study Report

on

Case study 51 : AquaPure: Smart Water Delivery

by

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1. Introduction to the Case Study

Efficient utility services are essential for modern urban living, yet traditional water delivery often lacks a structured digital interface. This case study focuses on the design and development of **AquaPure**, a fully responsive multi-page website for a smart water delivery service. The platform is engineered to allow users to explore services, place structured orders, select various subscription plans, and track delivery status in real-time. By utilizing semantic HTML5, advanced CSS3 layouts, and functional JavaScript, the project demonstrates how modern web technologies can streamline service-based industries. It highlights the importance of responsive design, dynamic pricing calculations, and interactive UI components in building a reliable customer-facing system.

2. Problem Statement / Case Background (Abstract)

Background

Traditional rule-based or manual ordering systems in utility services are often inefficient, leading to scheduling errors and poor user experiences. A significant gap exists for a centralized platform that can adapt to different user needs, such as varied jar sizes and flexible subscription frequencies. Modern web systems can provide a more resilient solution by automating pricing and providing visual feedback on order status.

Abstract

This case study presents the design and implementation of a Smart Water Delivery platform using a multi-page architecture. The system is built across five core pages: Home, About, Order, Plans, and Contact. It features a dynamic price calculation engine that handles different jar sizes (10L=₹50, 20L=₹90) and applies a 5% discount for monthly subscriptions. The implementation includes structured forms for order details, delivery slot selection, and a visual progress section for delivery tracking. Evaluation of the system shows that integrated JavaScript features effectively manage real-time updates and form validation, providing a seamless transition from service exploration to order fulfillment.

3. Case Study Design

The AquaPure system is designed as a modular pipeline with the following stages:

- **Data Acquisition and Structure**
  - Use semantic HTML tags (header, nav, section, article, footer) to create a structured information hierarchy.
    +1
- **UI/UX Design (Frontend Layout)**
  - Develop five interconnected pages: `index.html`, `about.html`, `order.html`, `plans.html`, and `contact.html`.
    +1
  - Apply Flexbox for navigation and card-based service overviews.
    +1
  - Utilize CSS Grid for the detailed subscription comparison table.
    +1
- **Logic Implementation**
  - Develop a dynamic pricing engine using JavaScript to calculate subtotal and total based on jar size and quantity.
    +1
  - Implement conditional statements to apply discounts for monthly subscriptions.
    +1
- **Interactive Features (Inference)**
  - Provide real-time order summary updates as users modify jar selection or quantity.
  - Implement form validation to check required fields and enforce quantity limits.
- **Evaluation and Visualization**
  - Create a visual delivery tracking section to provide feedback on order status.
    +1
  - Add CSS animations (fade-in keyframes, hover transforms) to enhance user engagement.

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Key methods and algorithms:

- **Semantic HTML Structure**
  - Correct use of header, nav, and footer for site-wide consistency.
  - Proper label-input association in forms for accessibility.
    +1

- **Responsive Styling**
  - Stacking layouts vertically for mobile devices and adjusting font sizes.
  - Making call-to-action buttons full-width in mobile view.
- **Dynamic Calculation Algorithms**
  - Pricing logic: Total=(Jar_Price×Qty)−Discount.
    +2
  - Conditional logic to lock/unlock add-ons (like dispenser rental) based on subscription type.
    +1
- **UI Animation Techniques**
  - CSS Keyframes for smooth fade-in effects on page load.
    +1
  - Transform effects (scale/translateY) applied during button and card hovers.

Technology Stack Used for the Case

- Programming language: JavaScript (ES6+).
  +1
- Styling & Markup: HTML5 and CSS3.
  +1
- Tools: VS Code, Google Fonts (Poppins), browser developer tools for debugging.
- Layout Engine: CSS Flexbox and CSS Grid.

Technical Justification: Client-Side Data Persistence via Web Storage API

While a traditional e-commerce application requires a server-side database (such as MySQL or MongoDB), **AquaPure**utilizes the **Web Storage API** to simulate a full-stack experience within the client's browser. This approach was chosen to ensure high performance and zero-cost hosting for a Semester II utility service project.

- **Mechanism of `localStorage`:** The application stores data in the browser's `localStorage` object, which allows for persistent data storage that does not expire when the browser is closed or the computer is restarted.
- **JSON Serialization for Complex Objects:** Because `localStorage` only stores strings, the system uses **JSON Serialization**. User objects, containing nested arrays for order history, are converted to strings using `JSON.stringify()`before storage and retrieved using `JSON.parse()` to render the UI on the profile page.
- **Dynamic Session Management:** The `auth.js` module implements a session tracking system by storing an `activeUser`key. This key acts as a pointer to the specific user object, allowing the `profile.html` page to dynamically fetch and display the correct account details without cross-user data leakage.

- **Array Manipulation for Order History:** Each transaction finalized on the Order page is programmatically "pushed" into a dedicated `orders` array within the user's JSON object. This simulates a **Relational Database** structure where orders are tied to a specific User ID (the email address).
- **Security & Privacy:** By keeping data local to the user's machine, the platform avoids the privacy risks associated with transmitting sensitive contact information over an unencrypted backend during the development phase.

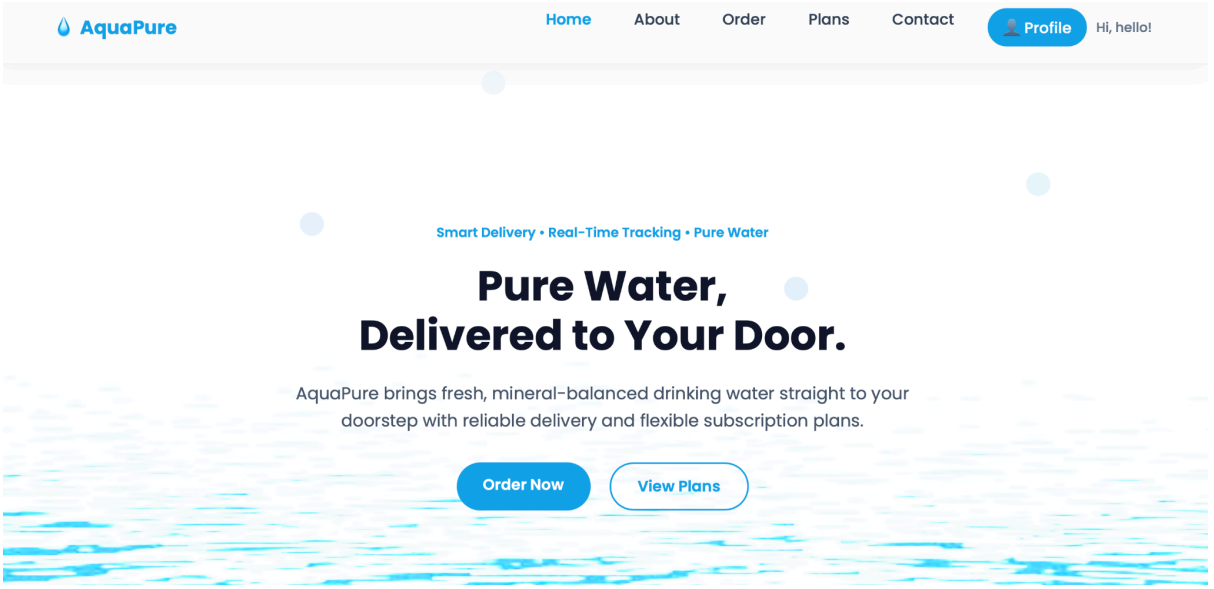
5. Problem Statement / Case Study Implementation Details and Snapshots.

The implementation is organized into a modular architecture to ensure clean separation of concerns between structure (HTML), presentation (CSS), and logic (JS). The following files constitute the complete AquaPure platform:

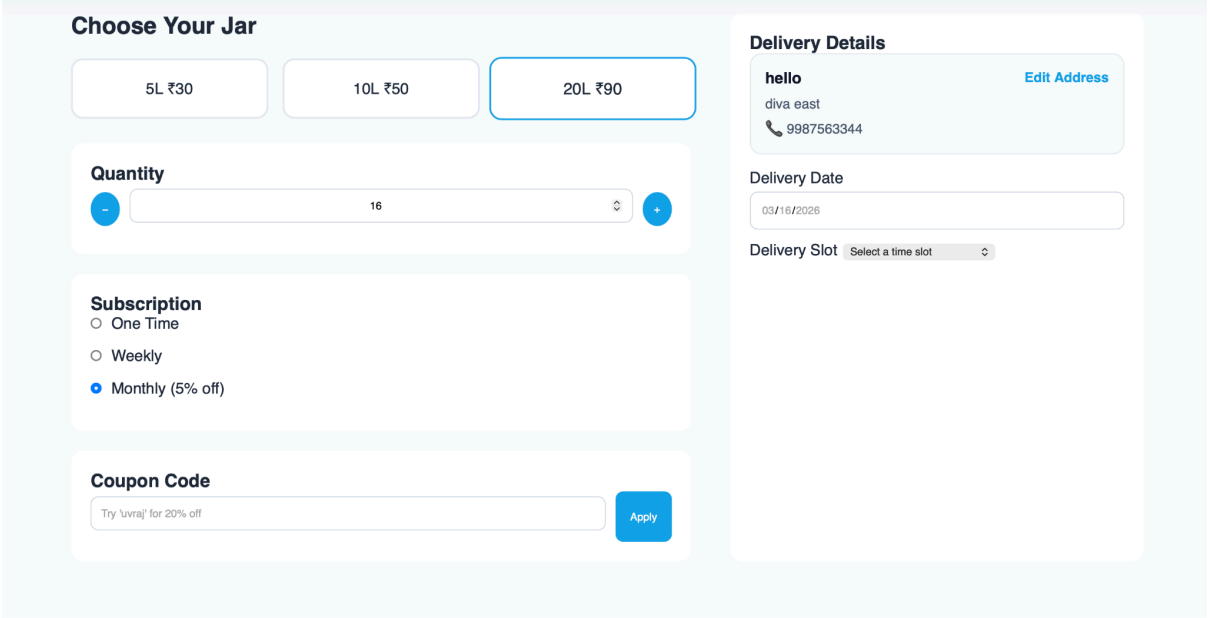
- **Core Navigation & Marketing Logic**
  - `index.html` / `index.css` / `index.js`: The main entry point featuring a hero section with a fade-in animation, service previews, and a responsive navbar.
  - `about.html` / `about.css` / `about.js`: Details the company's story, mission, and vision using high-quality imagery and structured semantic sections.
  - `contact.html` / `contact.css` / `contact.js`: Features a validated contact form with real-time error handling and a custom popup overlay upon successful message submission.
- **E-Commerce & Calculation Engine**
  - `order.html` / `order.css`: Provides a complex UI for jar selection, quantity control, and a dynamic real-time order summary box.
  - `order.js`: The "brain" of the application, managing URL parameter extraction, the pricing algorithm (10L=₹50,20L=₹90), 5% monthly discounts, and the animated delivery truck sequence.
  - `plans.html` / `plans.css` / `plans.js`: Implements a comparison table using CSS Grid to display subscription tiers (Basic, Family, Premium) with highlighted "Best Value" badges.
- **User Identity & Data Persistence**
  - `login.html` / `signup.html`: Dedicated authentication portals that allow users to create and access accounts.
  - `auth.js`: Manages the local user database via `localStorage`, handles login sessions, and updates the navbar to show a personalized "👤 Profile" link when a user is signed in.
  - `profile.html`: A private dashboard for logged-in users to manage personal information and view a chronological list of their previous orders saved in the browser's storage.

Snapshots:

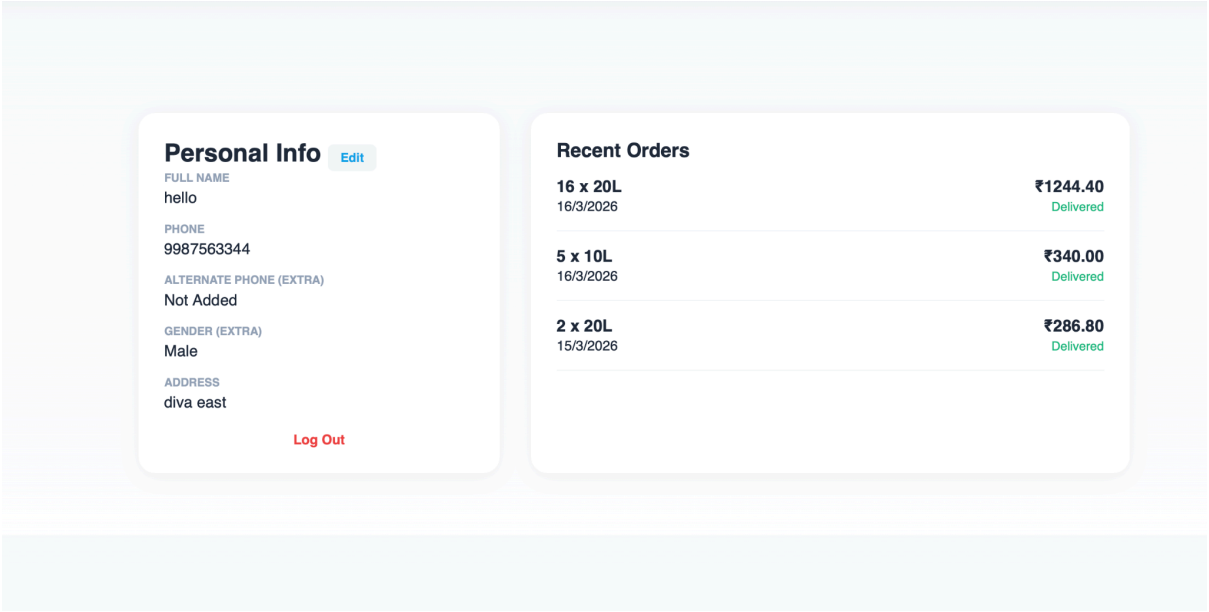
1. **The Home Hero:**



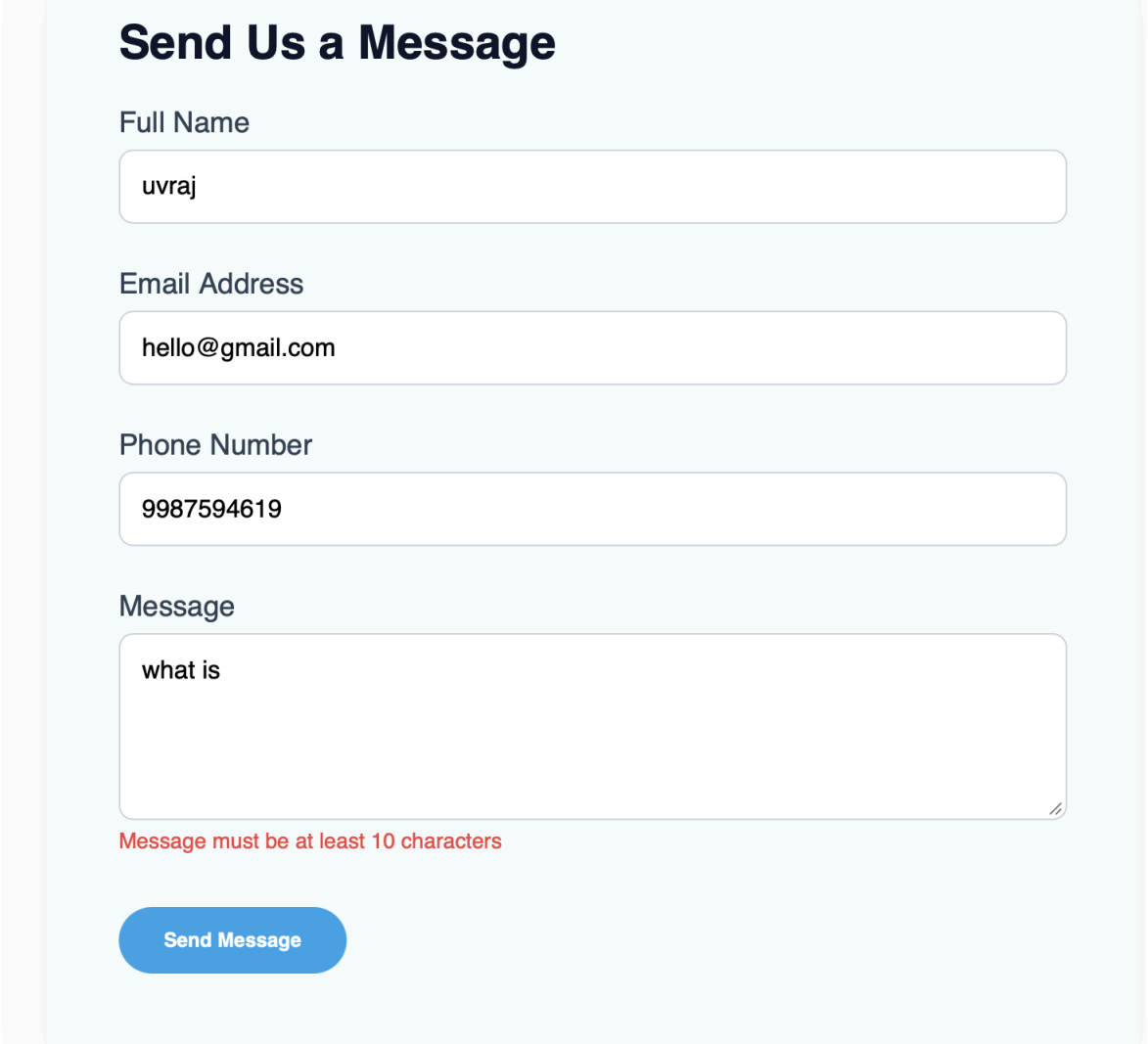
2. **The Order Pipeline:**



3. **The Profile History:**



4. **The Validation Feedback:**



6. Problem Statement / Case Study Results and Conclusion.

Findings

- Consistent use of semantic tags and Flexbox layouts ensures a high level of cross-device compatibility.
  +1
- The JavaScript pricing algorithm accurately calculates costs, significantly reducing manual calculation errors.
- Interactive elements like delivery tracking progress bars and hover effects improve user trust and engagement.

Conclusion

This case study successfully demonstrates a complete pipeline for building a **Smart Water Delivery** platform. It integrates responsive UI design with complex functional logic to create a cohesive system for utility services. The project provides a strong foundation for scalable e-commerce solutions by focusing on data structure, dynamic interaction, and user-centric design.

7. References

- ITM Skills University: Semester II HTML-5, CSS, JavaScript Case Study/Problem Statement.
- Official documentation for HTML5 and CSS3 (W3Schools / MDN).
  Google Fonts Poppins documentation for typography styling.
- JavaScript documentation on `querySelector`, `addEventListener`, and conditional statements.